

SARMAD SIDDIQUE

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PROFILE SUMMARY

MERN & AI Engineer with 3+ years of full-stack development experience and 1 year of hands-on AI engineering experience building scalable web applications and AI-powered healthcare solutions. Skilled in React.js, Next.js, TypeScript, Node.js, Express.js, MongoDB, REST APIs, and modern AI technologies including OpenAI APIs, LLMs, Prompt Engineering, RAG, Embeddings, Vector Databases, Tool Calling, and Agentic AI workflows. Currently building AI-powered healthcare agents, including Medical Scribe/Cortex, and contributing to the development of a Laboratory Information System (LIS) from scratch with Molecular Diagnostics, clinical workflow automation, and intelligent healthcare solutions.

TECHNICAL SKILL

- **Frontend:** React.js, Next.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS
- **Backend:** Node.js, Express.js, REST APIs, GraphQL APIs, JWT Authentication, Middleware Development
- **Databases:** MongoDB, Mongoose, Firebase, Vector Databases
- **AI & LLM:** OpenAI API, LLM Integration, Prompt Engineering, Agentic AI, Retrieval-Augmented Generation (RAG), Tool Calling, Embeddings, Vector Search, Whisper API, Structured Output Generation
- **Healthcare:** Laboratory Information System (LIS), Medical Scribing, SOAP Note Generation, Clinical Workflow Automation, Molecular Diagnostics
- **Tools:** Git, GitHub, Docker, GraphQL, Postman, npm, ESLint, Prettier

PROFESSIONAL EXPERIENCE

Texas Digital Hub - Lahore

MERN Stack & AI Engineer | June 2025 – Present

- Building and maintaining a Laboratory Information System (LIS) from scratch using React.js, TypeScript, Node.js, Express.js, and MongoDB.
- Developing AI-powered healthcare agents, including Medical Scribe/Cortex, using OpenAI APIs, LLMs, Prompt Engineering, Tool Calling, and Retrieval-Augmented Generation (RAG).
- Designing intelligent clinical workflows that combine patient history, laboratory data, AI reasoning, and structured SOAP note generation for physician review.
- Developed advanced Molecular Diagnostics modules including PCR Plate Mapping, Molecular Panel Management, Rules Engine, Sample Assignment, Analyzer Integration, and laboratory automation workflows.
- Built scalable REST APIs and reusable frontend architecture for clinics, providers, patients, laboratory tests, insurance, medical records, and reporting systems.
- Optimized application performance, reusable component architecture, and enterprise healthcare dashboards for improved scalability and maintainability.
- Collaborated with cross-functional engineering teams in Agile environments to deliver secure and production-ready healthcare solutions.

Ultraship TMS (Remote-US-Based)

Junior Software Engineer | Dec 2024 – Jun 2025

- Developed enterprise logistics software using Next.js, TypeScript, GraphQL, and modern frontend architecture.
- Built scalable modules including Masters Management, Calendar Scheduling, Email Marketing, Shipment Workflows, and operational dashboards.
- Integrated GraphQL APIs and developed reusable UI components to improve maintainability and development efficiency.
- Collaborated with distributed US-based engineering teams following Agile development practices.
- Improved application performance through modular architecture, reusable components, and TypeScript best practices.

Utecho - Faisalabad

React Developer | April 2023 – Nov 2024

- Led frontend development using React.js, Material UI, Ant Design, and Tailwind CSS.
- Assisted in backend API integration and database communication using Node.js and Express.js.
- Developed reusable UI components and integrated secure RESTful APIs.
- Worked closely with backend teams to ensure smooth frontend-backend communication.
- Participated in debugging, testing, Git-based version control, and deployment activities.
- Improved application performance and UI consistency across multiple projects.

Xavier-tech - Faisalabad

Frontend Developer | Feb 2023 – April 2023

- Developed responsive web pages and reusable React components.
- Converted UI/UX designs into functional frontend interfaces.
- Improved responsiveness, accessibility, and cross-browser compatibility.
- Collaborated with senior developers to enhance frontend performance and code quality.

PROJECTS

UltraShip TMS (Enterprise Logistics Platform)

- Developed enterprise logistics solutions using Next.js, TypeScript, GraphQL, Node.js, and modern frontend architecture.
- Built reusable UI components, integrated GraphQL APIs, and developed modules including Masters, Calendar, Email Marketing, and Shipment Management.
- Improved application performance and maintainability through modular architecture, reusable components, and close collaboration with distributed US-based engineering teams.

Laboratory Information System (LIS)

- Built a full-scale healthcare platform supporting clinics, providers, patients, insurance, laboratory testing, reporting, and operational workflows.
- Developed Molecular Diagnostics modules including PCR Plate Mapping, Molecular Panels, Rules Engine, Analyzer Integration, Sample Assignment, and laboratory automation
- Designed scalable frontend architecture and optimized enterprise healthcare workflows for laboratory operations.

Medical Scribe / Cortex AI

- Building an AI-powered healthcare agent that converts doctor-patient conversations into structured SOAP notes.
- Integrated OpenAI APIs, LLMs, Tool Calling, Retrieval-Augmented Generation (RAG), and patient clinical history retrieval.
- Designed Agentic AI workflows that combine clinical context, laboratory data, and AI reasoning to assist healthcare professionals with medical documentation.

EDUCATION

Government College University

Faisalabad, Pakistan

Bachelor of Information Technology

Sept 2020 – June 2024

ADDITIONAL INFORMATION

- 3+ years of experience building scalable MERN applications.
- 1 year of hands-on experience building AI-powered applications using OpenAI APIs, LLMs, RAG, and Agentic AI workflows.
- Experience integrating AI into production healthcare applications including Medical Scribe and Laboratory Information Systems (LIS).
- Strong understanding of scalable React architecture, REST APIs, backend systems, and healthcare workflow automation.
- Passionate about AI, Healthcare Technology, and Intelligent Software Systems.